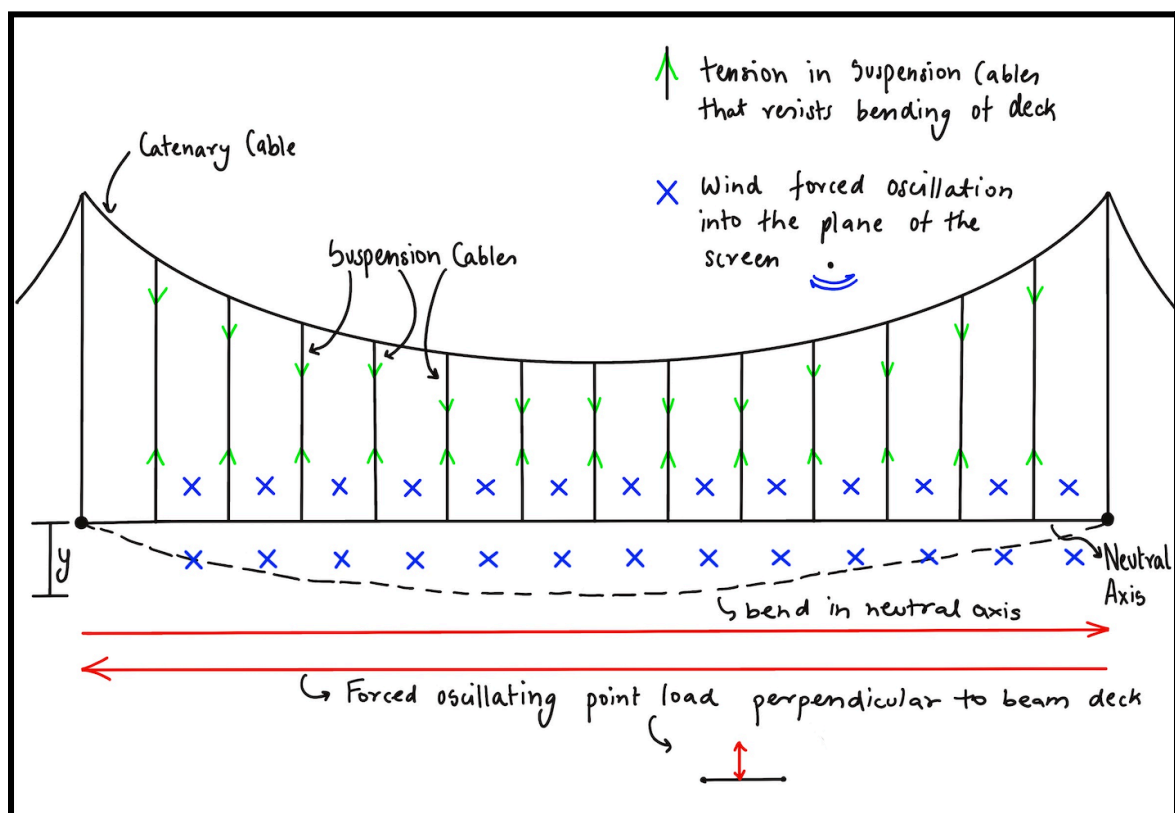


Research question- exploring a suspension bridge under complex dynamic spatial loads modelled by Catenary Curve Equation, Euler-Bernoulli Curve Equation and Lissajous oscillation curves for varying frequencies.

My exploration is based on a thought that prompted me to find the effect of spatial oscillations on a suspension bridge. After watching a video and learning about lissajous curves i realised that these curves give me dimensional frequencies and the period of oscillation in that dimension. Hence to spatially determine the effect on a beam deck suspended by a catenary curve with vertical suspension subjected to lissajous oscillations in the x and y direction I determined the following master differential equation. The purpose of this modelling is to depict more realistic load expectations that a bridge is subjected to. By using Lissajous curve equations we can incorporate any number of angular frequencies in any direction and analyse its effect on the vertical displacement of the beam deck.

Figure 1 below depicts the situation that is being modelled. The oscillating force acts in the plane going into the screen and parallel to the screen, perpendicular to the beam deck's neutral axis.



Master differential equation

$$EI \frac{d^4 y}{dx^4} + T_v \frac{d^2 y}{dx^2} + \rho A \frac{d^2 y}{dt^2} = \lambda g \cosh\left(\frac{x}{a}\right) + \frac{\sigma T_v}{L} + A \sin(w_x t) + B \cos(w_y t)$$

Dividing by EI we get

$$\frac{d^4 y}{dx^4} + \frac{T_v}{EI} \frac{d^2 y}{dx^2} + \frac{\rho A}{EI} \frac{d^2 y}{dt^2} = \frac{\lambda g}{EI} \cosh\left(\frac{x}{a}\right) + \frac{\sigma T_v}{LEI} + \frac{A}{EI} \sin(w_x t) + \frac{B}{EI} \cos(w_y t)$$

Hence

$$\frac{d^4 y}{dx^4} + \alpha \frac{d^2 y}{dx^2} + \beta \frac{d^2 y}{dt^2} = \frac{\lambda g}{EI} \cosh\left(\frac{x}{a}\right) + \frac{\sigma T_v}{LEI} + \frac{A}{EI} \sin(w_x t) + \frac{B}{EI} \cos(w_y t)$$

Where:

y is the vertical deflection of the beam at position x

$\frac{d^4 y}{dx^4}$ gives the behaviour of the beam's bend subjected to external forces where y is the

displacement function with respect to x . In beam theory the fourth derivative describes how the neutral axis's bending moment varies across the length of the beam.

E is young modulus of the beam's material which gives the materials stiffness

I is the moment of inertia of the beam

T_v is the tension in the vertical suspension cables that connect the catenary cable to the beam.

This tension helps restore the beam return to its equilibrium position and resists deflection

λ is mass per unit length of the catenary cable

g is acceleration due to gravity

σ is a scaling factor for the placement of the vertical suspensions

L is the length of the beam deck

A is the amplitude of the oscillatory load

w_x and w_y = angular frequencies in the x and y direction

$$a = \frac{T_h}{\lambda g}$$

$$\alpha = \frac{T_v}{EI}$$

$$\beta = \frac{\rho A}{EI}$$

In order to solve this equation we need to assume the solution to be of the following form:

$$y(x, t) = X(x)T(t) \text{--- (1)}$$

Where:

$X(x)$ - is a function that models the spatial variation of the beam's vertical displacement
(describes its bent shape).

$T(t)$ - is a function that models the time dependent oscillations.

Homogeneous and specific solution method:

The general solution to a non homogeneous differential equation is the sum of the homogeneous solution considered at y_c and a particular solution for each different static and dynamic load applied y_p . This principle known as the superposition principle is used in the following solution

$$y = y_c + y_p \text{ --- (2)}$$

In order to find y_c we will first assume that the load on the beam bridge is 0 (forming a homogeneous equation).

$$\frac{d^4 y}{dx^4} + \alpha \frac{d^2 y}{dx^2} + \beta \frac{d^2 y}{dt^2} = 0 \text{ --- (3)}$$

However the existence of the $\beta \frac{d^2 y}{dt^2}$ term makes this a partial differential equation. Hence

considering the assumption made in equation 1 we will separate the variables first by substituting

(1) in (3). Therefore:

$$\frac{d^4 X(x)T(t)}{dx^4} + \alpha \frac{d^2 X(x)T(t)}{dx^2} + \beta \frac{d^2 X(x)T(t)}{dt^2} = 0$$

$$X(x)'''' T(t) + \alpha X(x)'' T(t) + \beta X(x) T(t)'' = 0$$

Dividing by $X(x)T(t)$ to separate the variables we get:

$$\frac{X(x)''''}{X(x)} + \alpha \frac{X(x)''}{X(x)} + \beta \frac{T(t)''}{T(t)} = 0$$

Since x and t are independent variables they must be equal to the same constant that we will consider to be λ . where $\lambda > 0$ in order to get the temporal function on the form

$$T(t)'' + \omega^2 T(t) = 0$$

Thus we can form two following ordinary differential equations:

For the spatial function $X(x)$:

$$\frac{X(x)''''}{X(x)} + \alpha \frac{X(x)''}{X(x)} = \lambda$$

$$X(x)'''' + \alpha X(x)'' - \lambda X(x) = 0 \text{ --- (4)}$$

For the temporal function $T(t)$:

$$\beta \frac{T(t)''}{T(t)} = -\lambda$$

$$T(t)'' + \frac{\lambda}{\beta} T(t) = 0$$

Considering equation (4)

Using the method of trial solution

Let's assume that $y_c = e^{kx}$

Substitute y_c in homogeneous differential equation

$$\frac{d^4}{dx^4}e^{kx} + \alpha \frac{d^2}{dx^2}e^{kx} = 0$$

$$\text{We get } k^4 e^{kx} + \alpha k^2 e^{kx} = 0$$

$$\text{Since } e^{kx} \neq 0$$

We can divide by e^{kx} to get

$$k^4 + \alpha k^2 = 0$$

Upon factorising we get

$$k^2(k^2 + \alpha) = 0$$

Solving for k we get the following roots

$$\text{double root } k = 0 \quad \text{oscillatory complex roots } k = \pm i\sqrt{\alpha}$$

For the double root $k = 0$ the solution to $y_c^{(1)} = C_1 + C_2x$ as:

multiplicity of roots in linear differential equations- for repeated roots the characteristic equation requires adding terms involving x multiplied by constant to account for linearity of independent solutions and therefore:

$$y = (A_1 + A_2x + A_3x^2 + \dots)x^n e^{kn} \text{ so for } k = 0$$

$$y = A_1 + A_2x$$

$$\text{For the complex roots } k = \pm i\sqrt{\alpha} \text{ the solution to } y_c^{(2)} = C_3 \cos(\sqrt{\alpha}x) + C_4 \sin(\sqrt{\alpha}x)$$

$$\text{This is given by Euler's formula : } e^{i\sqrt{\alpha}x} = \cos(\sqrt{\alpha}x) + i\sin(\sqrt{\alpha}x)$$

Therefore the general solution is given by $y_c = y_c^{(1)} + y_c^{(2)}$

$$y_c = C_1 + C_2 x + C_3 \cos(\sqrt{\alpha x}) + C_4 \sin(\sqrt{\alpha x})$$

Now to solve for the particular solution of the equation:

The particular equation is as follows

$$\frac{d^4 y_p}{dx^4} + \alpha \frac{d^2 y_p}{dx^2} = \frac{\lambda g}{EI} \cosh\left(\frac{x}{a}\right) + \frac{\sigma T_v}{LEI} + \frac{A}{EI} \sin(w_x t) + \frac{B}{EI} \cos(w_y t)$$

Using the method of undetermined coefficients we will solve each term part of $q(x)$ to find the particular solution y_p where

$$1) \frac{\lambda g}{EI} \cosh\left(\frac{x}{a}\right)$$

$$2) \frac{\sigma T_v}{LEI}$$

$$3) \frac{A}{EI} \sin(w_x t) + \frac{B}{EI} \cos(w_y t)$$

$$\text{And } y_p = y_{p1} + y_{p2} + y_{p3}$$

Solving for catenary curve load $\cosh(\frac{x}{a})$

Let the particular solution be assumed to be in the form $y_{p1} = A \cosh(\frac{x}{a})$

$$- \frac{d^2}{dx^2} A \cosh\left(\frac{x}{a}\right) = \frac{A}{a^2} \cosh\left(\frac{x}{a}\right)$$

$$- \frac{d^4}{dx^4} A \cosh\left(\frac{x}{a}\right) = \frac{A}{a^4} \cosh\left(\frac{x}{a}\right)$$

Substitute into master equation considering $q(x) = q_{\text{catenary}}(x)$

$$\frac{A}{a^4} \cosh\left(\frac{x}{a}\right) + \alpha \frac{A}{a^2} \cosh\left(\frac{x}{a}\right) = \frac{\lambda g}{EI} \cosh\left(\frac{x}{a}\right)$$

Factorise $\cosh\left(\frac{x}{a}\right)$

$$A\left(\frac{1}{a^4} + \frac{\alpha}{a^2}\right) = \frac{\lambda g}{EI}$$

$$\text{Solving for } A = \frac{\lambda g}{EI\left(\frac{1}{a^4} + \frac{\alpha}{a^2}\right)}$$

$$\text{Therefore particular solution } y_{p1} = \frac{\lambda g}{EI\left(\frac{1}{a^4} + \frac{\alpha}{a^2}\right)} \cosh\left(\frac{x}{a}\right)$$

Solving for constant load due to vertical suspension strings $\frac{\sigma T_v}{LEI}$

Let the particular solution be assumed to be in the form $y_{p2} = Bx^4 + \frac{\sigma T_v}{LEI}$

- ***Since $B\frac{\sigma T_v}{LEI}$ is a constant***

$$- \frac{d^2}{dx^2} B_1 x^4 = 2B_1 x^2$$

$$- \frac{d^4}{dx^4} B \frac{\sigma T_v}{LEI} = 4B_1$$

$$4B + \alpha(2Bx^2) = Bx^4 + \frac{\sigma T_v}{LEI}$$

Comparing coefficients we get

$$B = \frac{\sigma T_v}{4LEI} \text{ since } B \neq 0$$

Therefore

$$y_{p2} = \frac{\sigma T_v}{4LEI} x^4 + \frac{\sigma T_v}{LEI}$$

Solving for oscillatory Load modelled using lissajous curve equations in two dimensions

$$\frac{A}{EI} \sin(w_x t) + \frac{B}{EI} \cos(w_y t)$$

Let the particular solution be assumed to be in the form $y_{p3} = \frac{A}{EI} \sin(w_x t) + \frac{B}{EI} \cos(w_y t)$

Since we are dealing with a fourth order partial differential equation we need to with respect to x and y while assuming that the other is constant.

Computing 2nd and 4th derivatives we get with respect to x

$$- \frac{d^2}{dx^2} D \sin(w_x x) \cos(w_y y) = - D w_x^2 \sin(w_x x) \cos(w_y y)$$

$$- \frac{d^4}{dx^4} \sin(w_x x) \cos(w_y y) = D w_x^4 \sin(w_x x) \cos(w_y y)$$

Computing 2nd and 4th derivatives we get with respect to y

$$- \frac{d^2}{dy^2} D \sin(w_x x) \cos(w_y y) = - D w_y^2 \sin(w_x x) \cos(w_y y)$$

$$- \frac{d^4}{dy^4} \sin(w_x x) \cos(w_y y) = D w_y^4 \sin(w_x x) \cos(w_y y)$$

Substitute into master equation considering $q(x) = q_{\text{lissajous oscillation}}(x)$ only

$$D w_x^4 \sin(w_x x) \cos(w_y y) + D w_y^4 \sin(w_x x) \cos(w_y y) - \alpha (D w_x^2 \sin(w_x x) \cos(w_y y) + D w_y^2 \sin(w_x x) \cos(w_y y)) = \frac{A}{EI} \sin(w_x x) \cos(w_y y)$$

$$(D w_x^4 + D w_y^4) \sin(w_x x) \cos(w_y y) - \alpha (D w_x^2 + D w_y^2) \sin(w_x x) \cos(w_y y) = \frac{A}{EI} \sin(w_x x) \cos(w_y y)$$

Compare coefficient of $\sin(w_x x) \cos(w_y y)$

$$(D w_x^4 + D w_y^4) - \alpha (D w_x^2 + D w_y^2) = \frac{A}{EI}$$

$$(D w_x^4 + D w_y^4)(1 - \alpha) = \frac{A}{EI}$$

$$D(w_x^4 + w_y^4) = \frac{A}{EI(1-\alpha)}$$

$$D = \frac{A}{EI(1-\alpha)(w_x^4 + w_y^4)}$$

$$y_{p3} = \frac{A}{EI(1-\alpha)(w_x^4 + w_y^4)} \sin(w_x x) \cos(w_y y)$$

Now upon culminating individual particular solutions to get y_p

$$y_p = y_{p1} + y_{p2} + y_{p3}$$

$$y_p = \frac{\lambda g}{EI(\frac{1}{a^4} + \frac{\alpha}{a^2})} \cosh(\frac{x}{a}) + \frac{\sigma T_v}{4LEI} x^4 + \frac{\sigma T_v}{LEI} + \frac{A}{EI(1-\alpha)(w_x^4 + w_y^4)} \sin(w_x x) \cos(w_y y)$$

Now we can find the general solution for y where

$$y = y_c + y_p$$

$$y = C_1 + C_2 x + C_3 \cos(\sqrt{\alpha} x) + C_4 \sin(\sqrt{\alpha} x) + \frac{\lambda g}{EI(\frac{1}{a^4} + \frac{\alpha}{a^2})} \cosh(\frac{x}{a}) + \frac{\sigma T_v}{4LEI} x^4 + \frac{\sigma T_v}{LEI} + \frac{A}{EI(1-\alpha)(w_x^4 + w_y^4)} \sin(w_x x) \cos(w_y y)$$

We can further apply boundary conditions to y and find the constant C_1 , C_2 , C_3 and C_4

We can also transform the term $\cos(w_y y)$ to be in x by considering the angle that winds blow at

and find the x -component of y and the angular frequency of the oscillation. In this similar

manner we can add multiple layers of oscillations in any direction and plane with respect to the beam deck as shown in **Figure 1** as the lissajous curves allow us to culminate these oscillations

to produces net oscillations in a certain direction to account for the beam decks stability and strength.

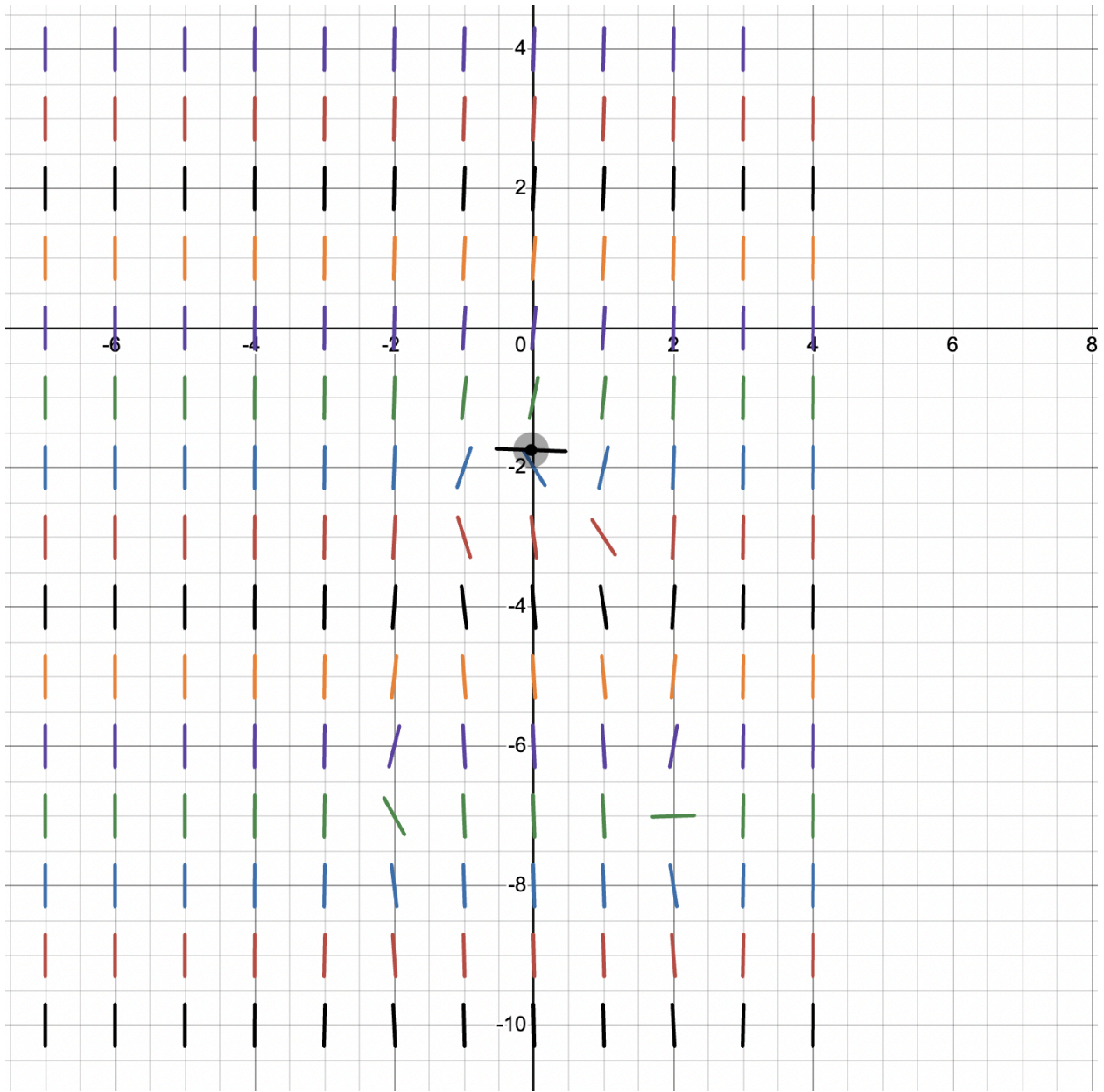


Image generated using Desmos and Euler's method to give slope fields and family of curves to fit modified 1st order differential equation for determined 4th order differential equation.

Some sources referred:

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Differential Equations in Engineering:

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Non homogeneous equation solving

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